

11. Random Variables

Transfer exploration seminar: Statistics and Data Science

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PSTAT 188

Summary: Basic Probability Theory

- Experiment, Sample space, Events
- Probability - Classical(Frequentist) Definition and Simulation based approach
- Probability Properties and Rules:
 1. The probability of an event A , denoted by $P(A)$, is a number between 0 and 1. $0 \leq P(A) \leq 1$
 2. For the sample space S , $P(S) = 1$
 3. $P(\emptyset) = 0$; $\emptyset$ is the null/empty set containing no elements.
 4. Complement: $P(A^c) = 1 - P(A)$
 5. Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 - **Special case:** If A and B are mutually exclusive events, that is, $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$
 6. Multiplication rule: $P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$, where $P(B|A)$ is the probability of event B given that event A happened.
 - **Special case:** If A and B are independent, ie $P(B|A) = P(B)$, then $P(A \cap B) = P(A) \times P(B)$

Next we will see. . .

Using series and integrals in probability calculation

Become familiar with the notation and concepts, the algebra will follow much more easily

Where do random variables come from?

Recall, for data(a sample) we said a variable can be

- Numerical - discrete or continuous
- Categorical - ordinal or nominal

When generalizing from a sample to the population

- There's always some uncertainty about the true distributions and relationships in the population
- Probability is the mathematical tool used to measure and express this uncertainty. (PSTAT 120A)
- **random variables** are the mathematical tool that allow us to incorporate uncertainty in the variables in our data.

What is a random variable?

A random variable X assigns a numerical value to each possible outcome (and event) of a random experiment.

Notation/Convention: Capital letters towards the end of the alphabet such as X, Y, Z are used to denote random variables

Corresponding lower case letters x, y, z are used to denote the observed outcomes (observed values/sample values).

Become familiar with the notation and concepts, the algebra will follow much more easily

Example

Experiment: Toss a fair coin

Outcome	H	T
Values: $X = x$	1	0

$$X = \begin{cases} 1 & \text{if coin lands heads} \\ 0 & \text{if coin lands tails} \end{cases}$$

Probability associated with values of a random variables

Instead of discussing probability of outcomes and events, we can focus on probability of values that a random variable takes.

Example Toss a fair coin

Outcome	H	T
Values: $X = x$	1	0
Probability: $P(X = x)$	1/2	1/2

Then, we can say that $P(X = 1) = 0.5$, i.e., X is equal to 1 with probability of 0.5

Probability distribution of a random variable

The probability distribution of a random variable specifies its possible values (i.e., its range) and their corresponding probabilities.

X = number of heads in a fair coin toss

As a table

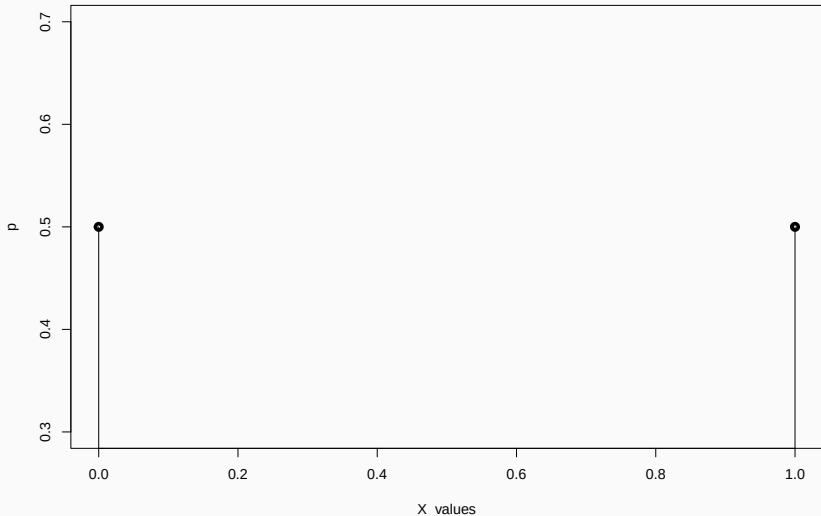
Values: $X = x$	0	1
Probability: $P(X = x)$	0.5	0.5

As a mathematical function

$$P(X = x) = \begin{cases} 1/2, & X = 1 \\ 1/2, & X = 0 \end{cases}$$

Probability distribution of a random variable

As a picture/graph/visualization



$$P(X = x) = \begin{cases} 1/2, & X = 1 \\ 1/2, & X = 0 \end{cases}$$

The total probability for the random variable is still 1 and all the probability rules we discussed still hold

$$\sum_{\text{all } x} P(X = x) = P(X = 0) + P(X = 1) = 1$$

Your turn : Theory : Getting ready for Las Vegas!

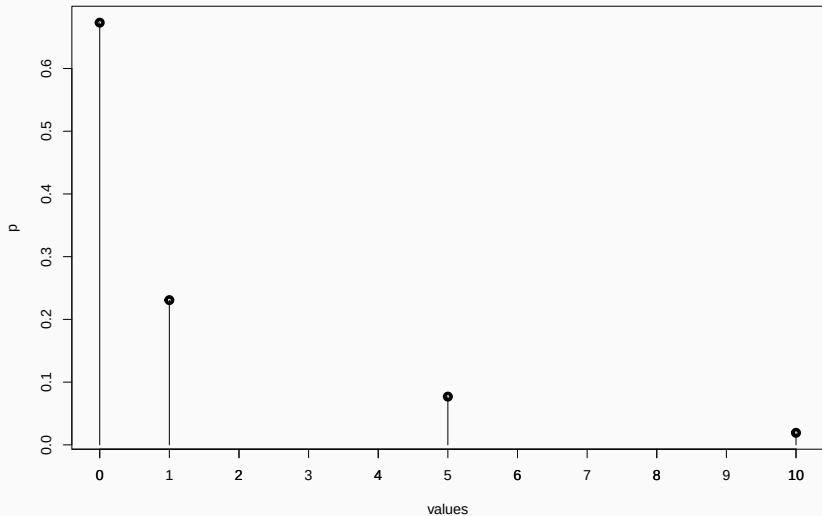
In a game of cards you win \$1 if you draw a heart, \$5 if you draw an ace (including the ace of hearts), \$10 if you draw the king of spades and nothing for any other card you draw. Write the probability mass function for your winnings.

- (What is the experiment?)
- What are the outcomes, sample space?
- What are you interested in counting?
- What is the random variable? it's values and probabilities?

The pmf for card game

Event	X	$P(X)$
Heart (not ace)	1	$\frac{12}{52} \approx 0.23$
Ace	5	$\frac{4}{52} \approx 0.08$
King of spades	10	$\frac{1}{52} \approx 0.02$
All else	0	$\frac{35}{52} \approx 0.67$
Total		1

The pmf for card game



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## [1] 0.67 0.23 0.08 0.02
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Would you bet on winning \$5 or \$10 at Las Vegas?

What is the probability that you win either \$5 or \$10?

$X = \$$ you win

$$\begin{aligned}P(X = 5 \text{ or } X = 10) &=_{\text{mutually exclusive}} P(X = 5) + P(X = 10) \quad (\text{addition rule}) \\&= 0.08 + 0.02 \\&= 0.1\end{aligned}$$

You have a 10% chance of winning either \$5 or \$10. Would you do it?

Study Skills: For practice Make up another event wrt this experiment and calculate its probability.

Be sure to write your solution out with explanations (how you use definitions, probability rules you use and how their usage is justified.)

Types of random variables

Recall: Types of variables in Statistical Data

For data(a sample) we said a variable can be

- Numerical - discrete or continuous
- Categorical - ordinal or nominal

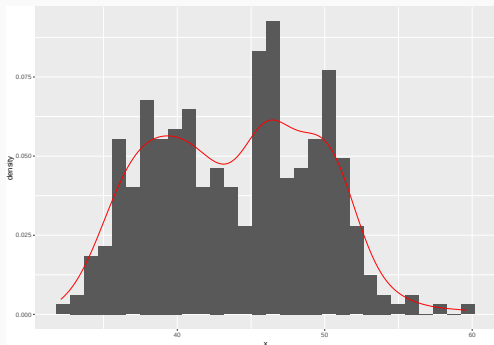
Types of random variables

A **discrete random variable** X has a finite or countably infinite number of possible outcomes. i.e we can *count* the number of outcomes

- The distribution function $P(X = x)$ is called a **probability mass function** (pmf)

A **continuous random variable** X takes all values in an interval of real numbers. i.e we can *measure* the outcomes.

From histograms to continuous distributions



If we use histograms to estimate continuous functions that describe all possible outcomes, we have created a probability density function.

Thus the probability for a continuous random variable can now be estimated by the **area under the probability density function/curve** $P(X < 40)$, $P(40 < X < 50)$ etc

Distribution of a continuous RV

- is specified by its **probability density function** (p.d.f.)
 - pdf gives the relative likelihood of the continuous random variable within the sample space.
 - can be represented by
 - a function $f(x)$, the density function or
 - its graph, the density curve
- the probabilities are given by the area under the graph between specified values.
 - If X is a continuous r.v., then $P(X = x) = 0$ for all values x .
- The total area under a density curve is always equal to 1.

Get used to the notation and the algebra/calculus will follow more easily.

Properties of the Probability Density Function (pdf) - $f(x)$

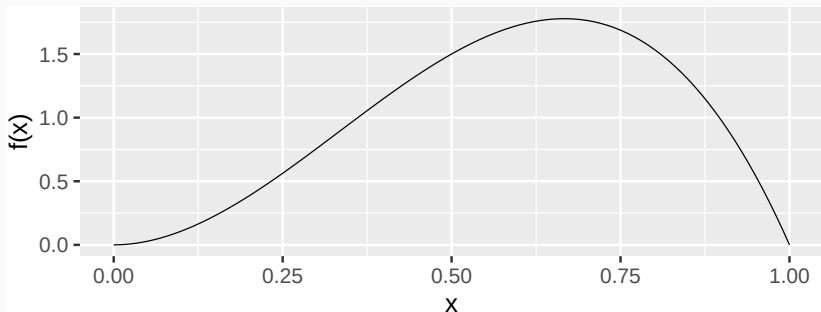
$$f(x) \geq 0 \text{ for all } x \in S$$

$$\int_{x \in S} f(x) dx = 1$$

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

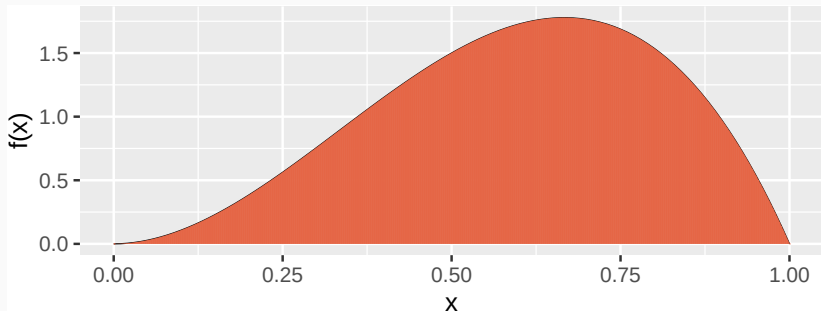
Example of continuous random variable

$$f(x) = \begin{cases} 12 x^2 (1 - x) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$



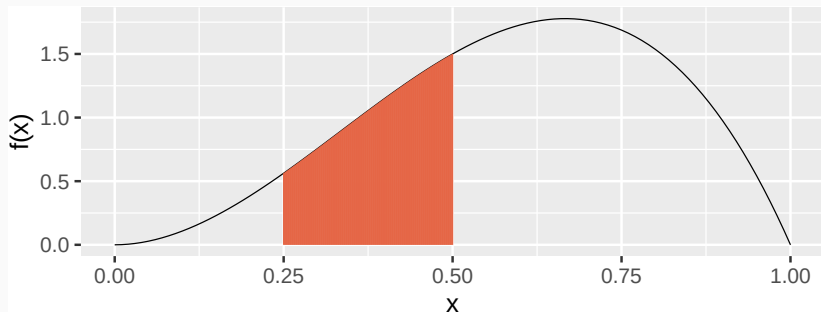
Area under the curve = 1

$$f(x) = 12(x^2)(1 - x), \quad 0 \leq x \leq 1$$



$$\begin{aligned} \int_{x \in S} f(x) dx &= \int_0^1 12(x^2)(1-x) dx \\ &= 12 \int_0^1 (x^2 - x^3) dx = 12 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 1 \end{aligned}$$

Probability is Area Under the Curve



$$\begin{aligned} P(0.25 < X < 0.50) &= \int_{0.25}^{0.50} 12(x^2)(1-x)dx = 12 \int_{0.25}^{0.50} (x^2 - x^3)dx \\ &= 12 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_{0.25}^{0.50} = 0.2617188 \end{aligned}$$

Descriptive summary for random variables

From sample measures to analogous population measures

For numerical data in a sample, we calculate **sample mean** $\bar{x}$ and **sample variance** s^2

We calculate analogous measures of center and spread for random variables (discrete, continuous).

The mathematical tools we use are **sums and series** for discrete random variables and **integrals** for continuous random variables.

Measure of center : Mean or average value or expected value of random variable X

	Notation	Formula
Sample	$\bar{x}$	$\frac{\sum_{i=1}^n x_i}{n} = \sum_{i=1}^n x_i \frac{1}{n}$
Discrete RV, pmf $P(X = x)$	$\mu = \bar{X} =$ $E(X)$	$\sum_{x_i \in S} x_i P(X = x_i)$
Continuous RV, pdf $f(x)$	$\mu = \bar{X} =$ $E(X)$	$\int_{x \in S} x f(x) dx$

Measure of spread: variance of random variable X

	Notation	Formula
Sample	s^2	$\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
Discrete Random Variable	$\sigma^2 =$ $Var(X)$	$\sum_{x_i \in S} (x_i - \bar{X})^2 P(X = x_i)$
Continuous Random Variable	$\sigma^2 =$ $Var(X)$	$\int_{x \in S} (x - \bar{X})^2 f(x) dx$

Summary:

- Random Variables: Discrete , Continuous
 - distribution function: pmf $P(X = x)$, pdf $f(x)$
 - probabilities : sums, series and integrals
 - Expected value: μ , $\bar{X}$, $E(X)$
 - Variance: σ^2 , $Var(X)$

Become familiar with the notation and concepts, the algebra will follow much more easily

Next we will see. . .

- Continue working with Random Variables
- Wrap up with a review.
- Hear your feedback about these modules.
- Discuss anything else you want.

Pre-reading: Math Review:

- Sum and series
- Integrals